

ASSIGNMENT REPORT

CUSTOMER CHURN PREDICTION AND SEGMENTATION SYSTEM

Course: Machine Learning

Dataset: Telco Customer Churn

Implementation: Python, Pandas, Scikit-learn

GitHub Repository: Add your repository link here

1. Title

Customer Churn Prediction and Segmentation System

2. Problem Statement

Customer churn is a major problem for service companies because losing existing customers can reduce revenue and increase the cost of acquiring new customers. The project develops a machine learning system that predicts whether a customer is likely to churn and groups customers with similar characteristics using K-Means clustering. The system uses customer demographic, service, contract, tenure, and billing information to support targeted retention decisions.

3. Objectives

- Load and understand a public customer churn dataset.
- Clean missing values and remove irrelevant attributes.
- Encode categorical attributes and scale numerical features where required.
- Identify high-risk customer groups and suggest retention actions.

4. Machine Learning Concepts Used

Concept	Usage in the Program
Pandas	Loads, cleans, transforms, and analyzes the customer dataset.
NumPy	Supports numerical arrays and data operations.
Data Preprocessing	Handles missing values, encoding, and feature preparation.
StandardScaler	Scales features for classification and clustering.
Train-Test Split	Separates data into training and testing sets using stratification.
Logistic Regression	Binary classifier for churn / non-churn prediction.
Decision Tree	Rule-based supervised classifier.
Gaussian Naive Bayes	Probabilistic supervised classifier.
Accuracy / Precision / Recall / F1	Measures classification performance.
Confusion Matrix	Shows true and false predictions for both classes.
K-Means	Groups customers into similar behavioral segments.

PCA	Reduces dimensions to two components for cluster visualization.
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5. Dataset Description

The project uses the Telco Customer Churn dataset. The dataset contains 7,043 customer records and 21 columns. It includes customer tenure, contract type, services, monthly charges, total charges, payment information, and the Churn target.

Dataset source used in the implementation:

https://raw.githubusercontent.com/Giskard-AI/examples/main/datasets/WA_Fn-UseC_-Telco-Customer-Churn.csv

6. Pseudocode

START

LOAD the Telco Customer Churn dataset

DISPLAY dataset shape and basic information

REMOVE customerID

CONVERT TotalCharges to numeric

IF TotalCharges contains missing values, REPLACE them with the median

ENCODE Churn: No = 0, Yes = 1

ONE-HOT ENCODE categorical attributes

SPLIT features and target

CREATE training and testing sets using stratified split

SCALE features using StandardScaler

TRAIN Logistic Regression

TRAIN Decision Tree

TRAIN Gaussian Naive Bayes

CALCULATE Accuracy, Precision, Recall and F1-score

DISPLAY confusion matrices and performance comparison

SELECT Logistic Regression as the best model based on F1-score

SCALE full feature matrix for clustering

CALCULATE K-Means inertia for K = 2 to 10

SELECT K = 4 from the elbow pattern

PERFORM K-Means clustering

APPLY PCA with two components

ANALYZE cluster size, churn rate, tenure and charges

IDENTIFY high-risk cluster

PROVIDE retention recommendations

END

7. Implementation – Data Loading and Preprocessing

```
import pandas as pd
import numpy as np

url = "https://raw.githubusercontent.com/Giskard-AI/examples/main/datasets/WA_Fn-UseC_-Telco-Customer-Churn.csv"
data = pd.read_csv(url)
print("Dataset Shape:", data.shape)

data = data.drop("customerID", axis=1)
data["TotalCharges"] = pd.to_numeric(data["TotalCharges"], errors="coerce")
data["TotalCharges"] = data["TotalCharges"].fillna(data["TotalCharges"].median())
data["Churn"] = data["Churn"].map({"No": 0, "Yes": 1})

X = pd.get_dummies(data.drop("Churn", axis=1), drop_first=True)
y = data["Churn"]
```

```
print("Features after encoding:", X.shape)
print("Missing values:", X.isnull().sum().sum())
```

The dataset contained 7,043 rows and 21 original columns. After removing customerID and encoding the categorical variables, the feature matrix contained 30 columns. Eleven missing TotalCharges values were handled using the median, leaving zero missing values.

8. Exploratory Data Analysis

Exploratory data analysis was performed before model training to understand churn distribution and important customer characteristics. The analysis included churn distribution, churn by contract type, monthly charges by churn status, and tenure by churn status.

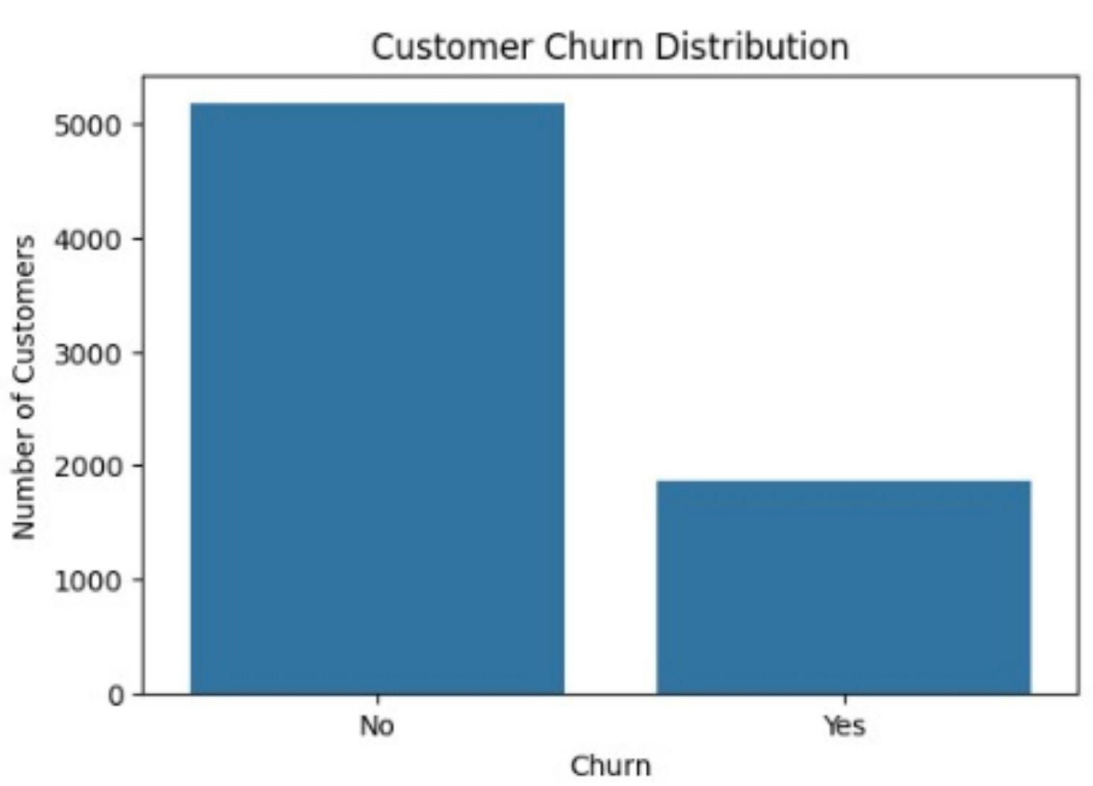


Figure 1: Customer Churn Distribution

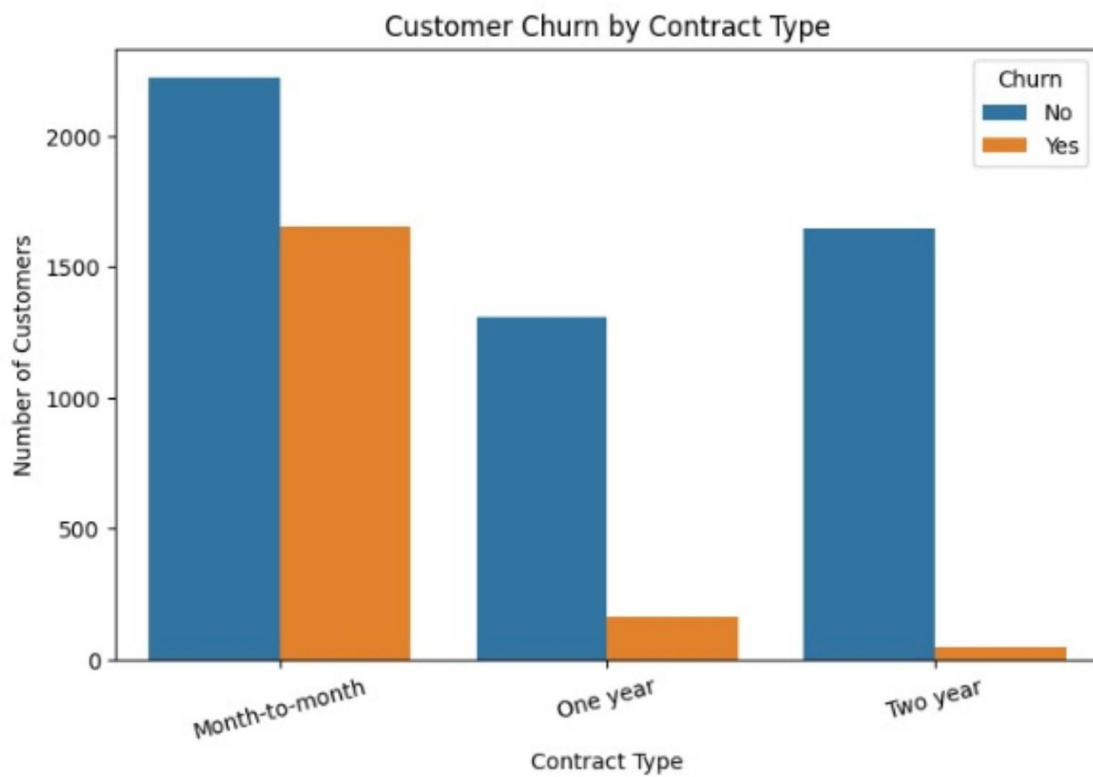


Figure 2: Customer Churn by Contract Type

8. Exploratory Data Analysis – Continued

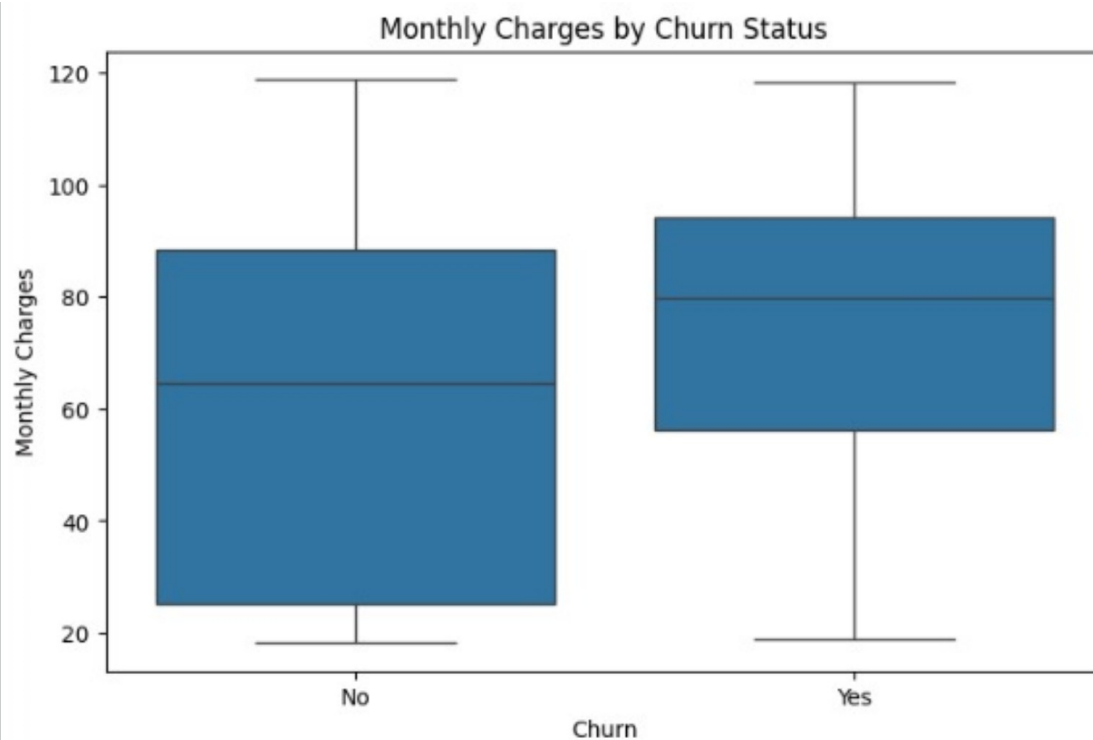


Figure 3: Monthly Charges by Churn Status

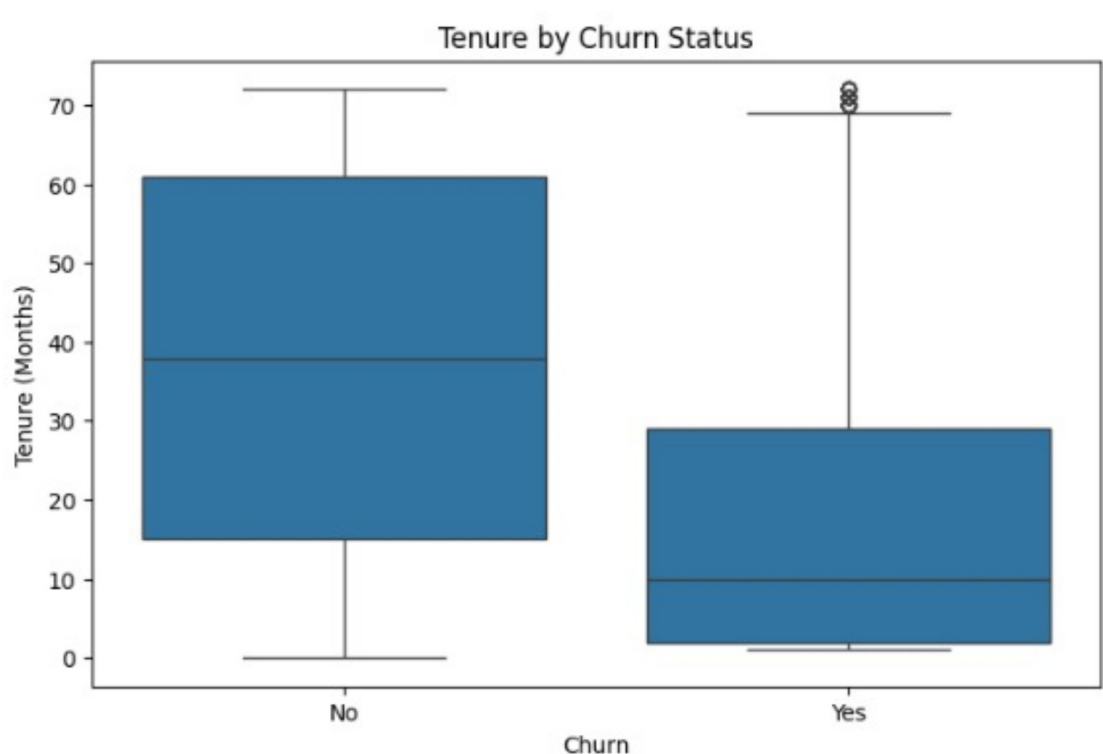


Figure 4: Tenure by Churn Status

The visualizations show that churn is not evenly distributed across customers. Contract type, monthly charges, and tenure provide useful signals for distinguishing customers who churn from those who remain.

9. Train-Test Split

The data was divided into training and testing sets using an 80:20 split with `random_state = 42` and stratification on the target variable. This preserves the approximate churn/non-churn class proportion in both sets.

Dataset	Samples	Features
Training set	5,634	30
Testing set	1,409	30

```
Training data shape: (5634, 30)
Testing data shape: (1409, 30)
Training target shape: (5634,)
Testing target shape: (1409,)

Data splitting and scaling completed successfully!
```

Figure 5: Train/Test Split Output

10. Supervised Machine Learning Models

Three supervised learning algorithms were trained as required by the assignment:

- Logistic Regression – provides a strong baseline for binary churn prediction.
- Decision Tree – learns decision rules from customer features.
- Gaussian Naive Bayes – estimates class probabilities using a probabilistic approach.

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.naive_bayes import GaussianNB
```

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=42),
    "Decision Tree": DecisionTreeClassifier(max_depth=5, random_state=42),
    "Gaussian Naive Bayes": GaussianNB()
}
```

```
for name, model in models.items():
    model.fit(X_train_scaled, y_train)
    y_pred = model.predict(X_test_scaled)
    # calculate evaluation metrics
```


11. Model Results

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8070	0.6584	0.5668	0.6092
Decision Tree	0.7942	0.6296	0.5455	0.5845
Gaussian Naive Bayes	0.6558	0.4269	0.8663	0.5719

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.806955	0.658385	0.566845	0.609195
1	Decision Tree	0.794180	0.629630	0.545455	0.584527
2	Naive Bayes	0.655784	0.426877	0.866310	0.571933

Figure 6: Model Results Output

Logistic Regression produced the highest accuracy, precision, and F1-score among the three tested models. Gaussian Naive Bayes produced the highest recall, meaning it detected more churners, but it also generated substantially more false positives.

12. Confusion Matrix

The confusion matrices show the number of correctly and incorrectly classified customers.

Model	TN	FP	FN	TP
Logistic Regression	925	110	162	212
Decision Tree	915	120	170	204
Gaussian Naive Bayes	600	435	50	324

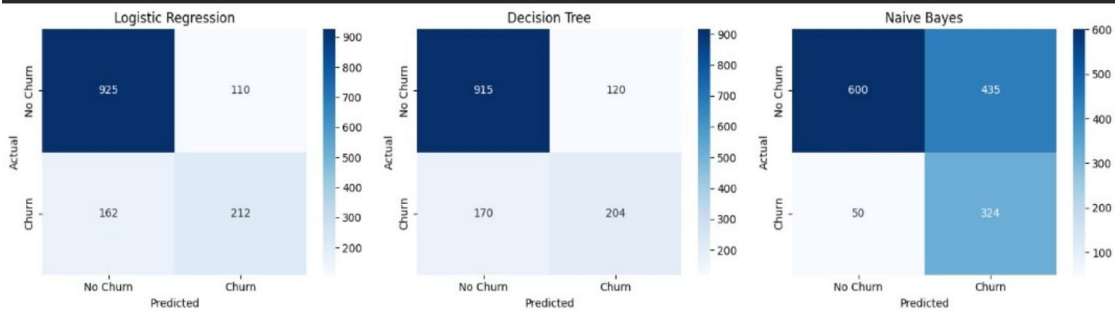


Figure 7: Confusion Matrices

Logistic Regression provides a more balanced error profile. Naive Bayes detects more actual churners but labels many non-churners as churners, producing 435 false positives.

13. Model Performance Comparison

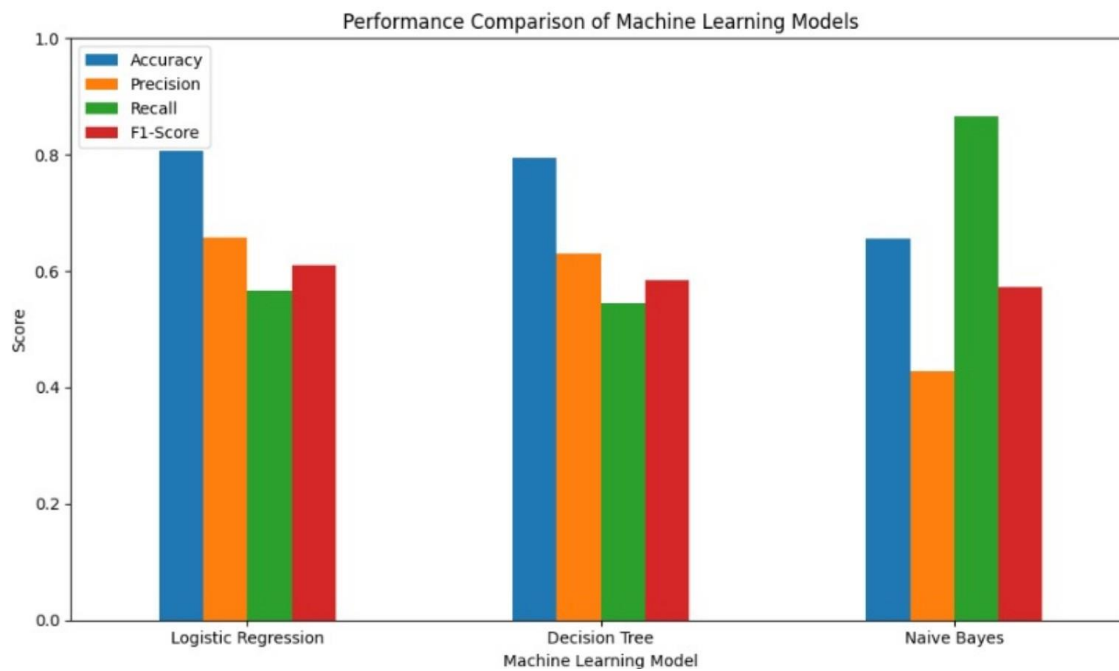


Figure 8: Performance Comparison

```
Best Performing Model:
Model          Logistic Regression
Accuracy              0.806955
Precision            0.658385
Recall               0.566845
F1-Score             0.609195
Name: 0, dtype: object
```

Figure 9: Best Model Output

The best model is Logistic Regression because it achieved the highest F1-score of 0.6092 and the highest accuracy of 0.8070. F1-score is useful here because churn prediction requires a balance between finding customers who churn and avoiding excessive false alarms.

14. K-Means Clustering

K-Means clustering was applied to the standardized customer feature matrix to identify groups of customers with similar characteristics. The elbow method was evaluated for K values from 2 to 10.

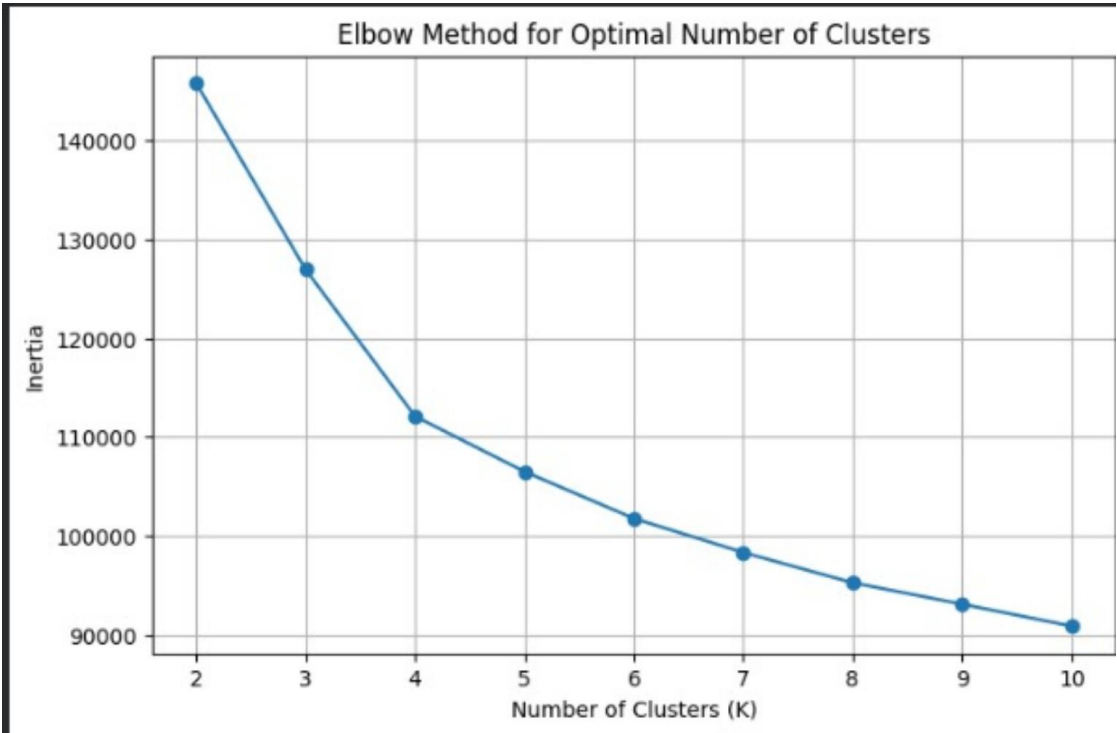


Figure 10: Elbow Method

Based on the visible elbow pattern, K = 4 was selected. K-Means was then trained using `n_clusters = 4`, `random_state = 42`, and `n_init = 10`.

15. Cluster Distribution

```
K-Means clustering completed successfully!

Number of customers in each cluster:
Cluster
0      682
1     2735
2     2100
3     1526
Name: count, dtype: int64
```

Figure 11: K-Means Cluster Counts

Cluster	Customers	Share
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0	682	9.68%
1	2,735	38.83%
2	2,100	29.82%
3	1,526	21.67%

Cluster labels are identifiers generated by K-Means and do not represent an inherent ranking. The clusters are interpreted using their customer size, churn rate, tenure, and billing characteristics.

16. PCA Visualization

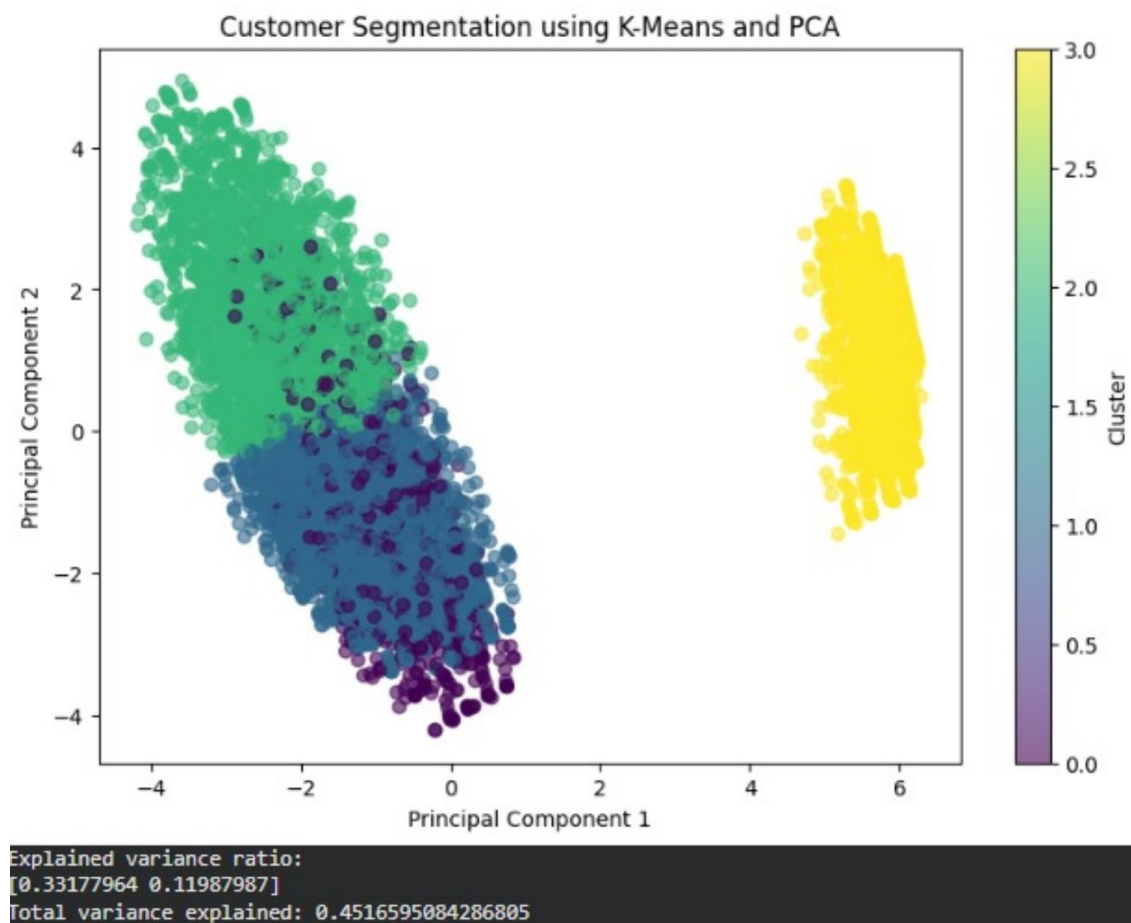


Figure 12: PCA Cluster Visualization

Principal Component Analysis was used to reduce the 30-dimensional feature space to two components for visualization. PC1 explains 33.18% of the variance and PC2 explains 11.99%, giving a combined explained variance of approximately 45.17%. The PCA plot provides a two-dimensional view of the customer segments.

17. Cluster Analysis

Cluster	Customers	Churned	Churn Rate	Avg Tenure	Avg Monthly	Avg Total
0	682	170	24.93%	31.74	42.03	1500.13
1	2,735	1,277	46.69%	15.54	73.99	1172.30
2	2,100	309	14.71%	55.83	91.87	5153.66
3	1,526	113	7.40%	30.55	21.08	668.10

```
Customer Segment Analysis:
  Customers  Churned  Churn_Rate  Avg_Tenure  Avg_Monthly_Charges  \
Cluster
0          682      170      24.93      31.74          42.03
1         2735     1277      46.69      15.54          73.99
2         2100      309      14.71      55.83          91.87
3         1526      113       7.40      30.55          21.08

      Avg_Total_Charges
Cluster
0          1500.13
1          1172.30
2          5153.66
3           668.10
```

Figure 13: Cluster Analysis Output

18. High-Risk Customer Segment

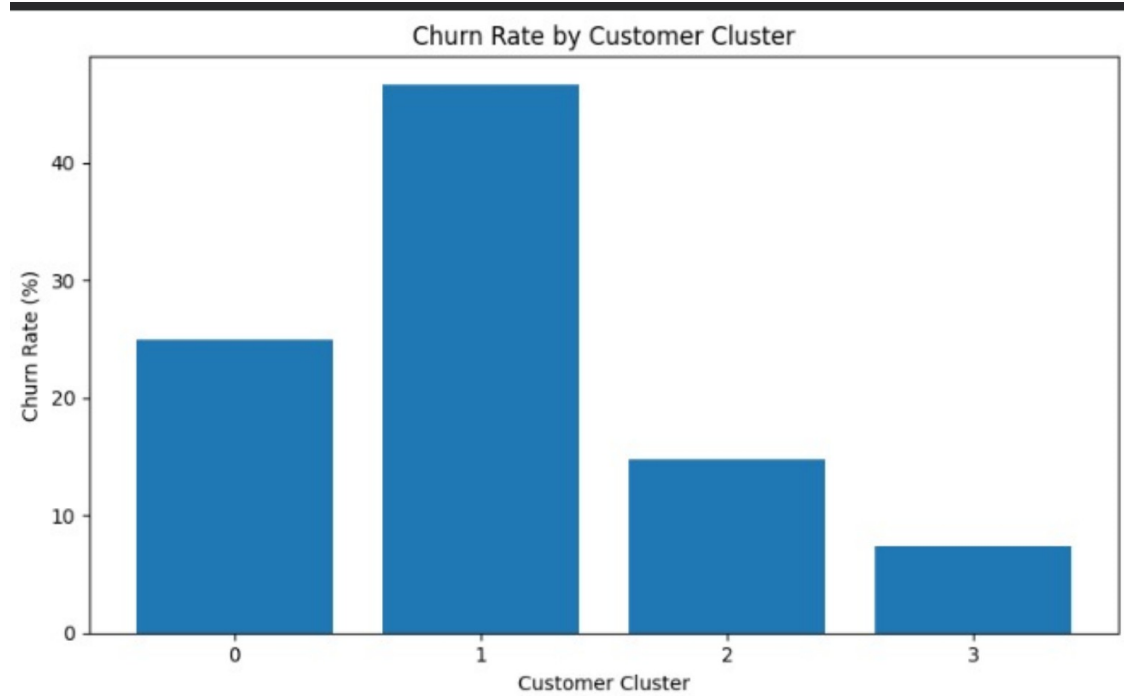


Figure 14: Churn Rate by Cluster

... FINAL CUSTOMER SEGMENT SUMMARY					
	Customers	Churned	Churn_Rate	Avg_Tenure	Avg_Monthly_Charges \
Cluster					
0	682	170	24.93	31.74	42.03
1	2735	1277	46.69	15.54	73.99
2	2100	309	14.71	55.83	91.87
3	1526	113	7.40	30.55	21.08
Avg_Total_Charges			Risk Level		
Cluster					
0	1500.13	Low/Medium Risk			
1	1172.30	High Risk			
2	5153.66	Low/Medium Risk			
3	668.10	Low/Medium Risk			

Figure 15: Final Customer Segment Summary

Cluster 1 is the highest-risk segment with a churn rate of 46.69%. It contains 2,735 customers, has an average tenure of 15.54 months, and an average monthly charge of 73.99. This combination suggests that early-tenure customers with relatively higher monthly charges deserve priority retention attention.

19. Retention Recommendations

Cluster	Risk / Profile	Recommended Action
0	Moderate churn; 24.93%	Targeted engagement, loyalty benefits, and responsive support.
1	High churn; 46.69%	Priority retention, early-tenure onboarding, plan review, targeted incentives, proactive support.
2	Low churn; 14.71%; long tenure	Protect high-value loyal customers with loyalty benefits and premium support.
3	Lowest churn; 7.40%; low monthly charges	Maintain service quality, low-cost loyalty offers, and referral engagement.

The recommendations are based on the observed cluster characteristics. They should be treated as business decision support rather than automatic actions; actual offers should be validated using cost, customer preferences, and company policy.

20. Test Cases and Validation

Test	Input / Condition	Expected Output	Validation
Dataset Loading	Read public CSV	7,043 rows loaded	Successful
Missing Values	11 TotalCharges values	Replace with median	0 missing values after handling
Encoding	Categorical attributes	Numeric feature matrix	30 features obtained
Train/Test Split	80:20, stratified	5,634 / 1,409	Successful
Model Training	3 classifiers	Predictions generated	Successful
Evaluation	Test predictions	Accuracy, Precision, Recall, F1	Successful
Confusion Matrix	Each classifier	2×2 matrix	Generated
Elbow Method	K = 2 to 10	Inertia values / elbow	K = 4 selected
K-Means	4 clusters	Cluster labels generated	Successful
PCA	2 components	2D cluster visualization	Generated

21. Result

The Customer Churn Prediction and Segmentation System was implemented successfully. Logistic Regression achieved the best overall classification performance with 80.70% accuracy and an F1-score of 60.92%. K-Means identified four customer segments, with Cluster 1 showing the highest churn rate of 46.69%. The system therefore combines prediction and segmentation to identify customers who may require targeted retention strategies.

22. Sample Output

The following screenshots show the main outputs generated during execution.

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.806955	0.658385	0.566845	0.609195
1	Decision Tree	0.794180	0.629630	0.545455	0.584527
2	Naive Bayes	0.655784	0.426877	0.866310	0.571933

Model evaluation output

... FINAL CUSTOMER SEGMENT SUMMARY					
Cluster	Customers	Churned	Churn_Rate	Avg_Tenure	Avg_Monthly_Charges \
0	682	170	24.93	31.74	42.03
1	2735	1277	46.69	15.54	73.99
2	2100	309	14.71	55.83	91.87
3	1526	113	7.40	30.55	21.08

Cluster	Avg_Total_Charges	Risk Level
0	1500.13	Low/Medium Risk
1	1172.30	High Risk
2	5153.66	Low/Medium Risk
3	668.10	Low/Medium Risk

Final customer segment summary

23. Implementation – Main Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

models = [
    LogisticRegression(max_iter=1000, random_state=42),
    DecisionTreeClassifier(max_depth=5, random_state=42),
    GaussianNB()
]
```



```

for model in models:
    model.fit(X_train_scaled, y_train)
    pred = model.predict(X_test_scaled)
    print(accuracy_score(y_test, pred))
    print(precision_score(y_test, pred))
    print(recall_score(y_test, pred))
    print(f1_score(y_test, pred))

# K-Means
cluster_scaler = StandardScaler()
X_cluster = cluster_scaler.fit_transform(X)
inertias = []
for k in range(2, 11):
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_cluster)
    inertias.append(km.inertia_)

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_cluster)

# PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_cluster)
print(pca.explained_variance_ratio_)

```

24. GitHub Link

Source Code Repository:

The final repository should contain the complete working notebook/source code, dataset reference, required dependencies, and any supporting files needed to reproduce the results.

25. Conclusion

The project demonstrates an end-to-end machine learning workflow for customer churn prediction and customer segmentation. Data preprocessing prepared the customer records for analysis, while

exploratory visualizations helped understand churn-related patterns. Three supervised models were compared, and Logistic Regression provided the best overall balance of performance. K-Means clustering identified four customer segments and revealed a high-risk group with a 46.69% churn rate. These results can be used to support targeted customer retention strategies.

26. References

1. Telco Customer Churn dataset, IBM/GitHub public dataset source.
2. Scikit-learn documentation – classification, preprocessing, clustering, and PCA.
3. Verbeke, W. et al. Customer churn prediction using logistic regression and decision trees.
4. Sagming, P. et al. Using topological data analysis and machine learning to predict customer churn, *Journal of Big Data*, 2024.
5. Recent research literature on explainable and machine-learning-based customer churn prediction, 2025–2026.